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L05 ML Data Preparation

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Reflection on Data Preparation

Before this lab, our group mainly saw data preparation as a cleaning step before model training. After working with the Titanic dataset, we realized that preprocessing requires more judgment than we expected. Each decision can change what a future model learns from the data, whether we handle missing values, convert categories into numbers, or scale numerical features. This lab also helped us connect preprocessing to our earlier Exploratory Data Analysis work because the problems we found during EDA helped guide our preprocessing choices. Instead of viewing these steps as separate tasks, we started to understand how they build on one another throughout the machine learning workflow and why mistakes in one stage can affect later results.

Handling the missing values in Age changed how we thought about incomplete data. The dataset contained 177 missing age values, so removing every row with a missing age would also remove a large amount of other useful information. We used the median because it allowed us to keep those rows while reducing the effect of unusually high or low ages on the replacement value (Tien). However, this task also made us realize that imputation cannot recover a passenger's real age because it only gives the model a reasonable estimate (Tien). This helped us understand that imputation does not completely "fix" missing data. Instead, it gives us a way to keep useful observations while accepting some uncertainty (Tien). It also showed us that even a simple preprocessing decision can involve trade-offs, which means we need to understand the data before choosing a method (Tien).

One-Hot Encoding taught us that converting categorical data into numbers takes more thought than simply assigning a number to each category. At first, using values like 1, 2, and 3 may seem easy, but those numbers can suggest an order that does not actually exist (Viktoriya). By converting Sex and Embarked into columns that use 0s and 1s, One-Hot Encoding gave the model a numerical representation without suggesting that one category ranks above another (Viktoriya). This made it clear that preprocessing can affect how a model understands the data, not just how the dataset looks (Viktoriya). Using `drop_first=True` also taught us that we can remove redundant information while still keeping enough information to represent the categories correctly (Viktoriya). This task made the role of encoding clearer because we could see that the goal involves preserving meaning while changing the format into something a model can use (Viktoriya).

Feature scaling added another layer to our understanding because Age and Fare originally had very different numerical ranges. `StandardScaler` placed both features on a similar scale while

keeping each value's position relative to the others (Zaid). This helped us see why some algorithms may respond differently when one feature contains much larger numbers, even though larger values do not make that feature more important (Zaid). At the same time, the Knowledge Check showed us that not every algorithm needs the same preparation because Decision Trees make splits based on feature values rather than relying on the numerical scale in the same way as some other models (Zaid). This challenged our earlier assumption that we should automatically use every preprocessing technique for every machine learning problem (Zaid). We now understand that our choice of model should guide the preprocessing steps we use, which connects data preparation more closely to model selection than we first expected (Zaid).

Additionally, the lab gave us a practical lesson about how connected each step of the workflow is. Some notebook cells did not run or appear correctly, so we had to correct them and check that each step still worked before moving forward (Tien). This problem showed us why working systematically matters because Task 2 depended on the earlier preparation, while Task 3 worked with the encoded DataFrame that Task 2 created (Tien, Viktoriya, and Zaid). Re-running cells and checking the outputs helped us see why reproducibility is important, especially when several people work on different parts of the same notebook (Tien). We connected this process to our previous EDA lab as well. EDA helped us find problems in the dataset, while this lab showed us how to deal with those problems before model training (Viktoriya). Seeing how each stage connects made the whole workflow feel less overwhelming because every step now has a clearer purpose and leads into the next one.

The lab raised several questions that we would like to explore in future machine learning tasks. We want to compare mean and median imputation for Age and see whether the choice changes model performance enough to make a difference (Tien). We also became curious about what happens when One-Hot Encoding handles a feature with many more categories, because creating too many new columns could make the dataset much larger and more difficult to work with (Viktoriya). Another question is how much feature scaling affects algorithms that rely strongly on numerical distances compared with models such as Decision Trees (Zaid). These questions showed us that successful preprocessing involves more than making the code run correctly. We need to test how our choices affect later results and decide whether each preprocessing step actually improves the model.

In conclusion, this lab changed how our group views data preparation within the machine learning workflow. We now see preprocessing as a series of decisions that shape what information a model receives rather than as a simple cleaning stage. Imputation, encoding, and scaling each solve different problems, and the best method depends on the data and the model we plan to use. The lab made the importance of careful checking and reproducibility clearer, especially when several preprocessing steps depend on one another. Most importantly, we learned to question why we use each technique, what trade-offs it may create, and how each choice can affect the next stage of the machine learning process.